



HAZOP Leadership training for GC

Chapter 3 – Introduction to HAZOP study

Technical Safety Service Division (Q-TS-TS), 1 - 3 November 2021

Table of content

HAZOP Leadership Training Course

- ☐ What is HAZOP?
- ☐ History of HAZOP
- ☐ How and Why HAZOP is Used
- ☐ Purpose & Objective of HAZOP
- ☐ HAZOPs Limitation
- ☐ Basic Features of a HAZOP

What is HAZOP ?

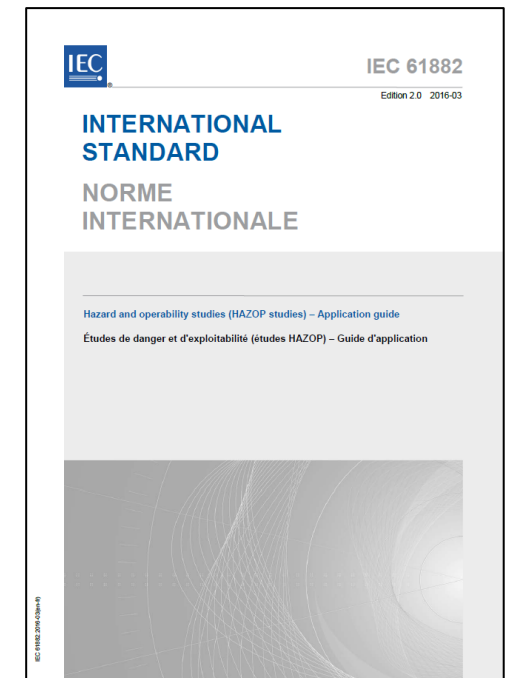
HAZOP Leadership Training Course

- ❑ Systematic technique to IDENTIFY potential hazards and operating problems
- ❑ A **qualitative technique** based on “*guide-words*” to help provoke thoughts about the way deviations from the intended operating conditions can lead to hazardous situations or operability problems
- ❑ Deviations developed through application of guide words to appropriate process parameters
- ❑ HAZOP is basically for safety
 - Hazards are the main concern
 - Operability problems degrade plant performance (product quality, production rate, profit)

History of HAZOP

HAZOP Leadership Training Course

- ❑ Initially developed by Dr. H G Lawley (1974) of ICI at Wilton based on early account by Elliott & Owen (1968).
- ❑ Subsequently C J Bullock and A J D Jennings from ChE Dept. Teesside Polytechnic under supervision of T A Kletz applied the method at higher institution (Post-graduate level).
- ❑ First guide: “A Guide to Hazard and Operability Studies”, ICI and Chemical Industries Associations Ltd.1977.



How and Why HAZOP is Used

HAZOP Leadership Training Course

- ☐ Identifies potential hazards , failures and operability problems.
- ☐ Most effective as a team effort of plant and process designers, operating personnel, control, instrumentation engineers and safety engineer.
- ☐ Identifies causes and the consequences of perceived mal – operations of equipment and associated operator interfaces in the context of the complete system
- ☐ Encourages creativity in design concept evaluation
- ☐ Results in fewer commissioning & operational problems and better informed personnel, thus confirming overall cost effectiveness improvement
- ☐ Suggests necessary changes to a system for eliminating or reducing the probability of operating deviations

How and Why HAZOP is Used – cont.

HAZOP Leadership Training Course

- ❑ Provides a necessary management tool in so far that it demonstrates to insurers & inspectors evidence of comprehensive thoroughness.
- ❑ HAZOP reports are an integral part of plant and safety records and are also applicable to design changes and plant modifications, thereby containing accountability for equipment and its associated human interface throughout the operating lifetime.
- ❑ Recommended as a principal method by professional institutions and legislators on the basis of proven capabilities for over 40 years.
- ❑ Used by most major companies handling and processing hazardous material, especially those where engineering practice involves elevated operating parameters (O&G, Refinery, Petrochem, Pharmaceuticals)

Purpose & Objective of HAZOP

HAZOP Leadership Training Course

- ❑ It emphasizes upon the operating integrity of a system, thereby leading methodically to most potential and detectable deviations which conceivably arise in the course of normal operating routine
 - including “stat-up” and “shut – down”
 - as well as steady-state operations.

- ❑ The objectives of HAZOP study are to:
 - Identify hazards and operability problems;
 - Assess the extent of the hazard or operability problems;
 - Identify the safeguard for each identified hazardous event;
 - Propose recommendations to prevent or mitigate the consequence; and
 - Serve as input to risk management activities.

- ❑ Not suitable for examining the high-level issues such as the details of layout or the quantification of risks
- ❑ HAZOP is not the means for defining engineering and procedural solutions for sources of hazards.
 - Check the rating of the individual flanges or individual vessel designs
 - Check rated values of electric cables
 - Check operating instructions
 - Company policies / emergency proceduree
- ❑ It is important to remember at all times that HAZOP is an identifying technique and not intended as a means of solving problems nor is the method intended to be used solely as an undisciplined means of searching for hazardous scenarios.
- ❑ The purpose of the HAZOP is not to challenge the design conditions or assumptions, although such challenges do arise a must be faced
- ❑ HAZOP is a time consuming, and shortcuts are not permitted.
- ❑ The system being studied should be well defined.
- ❑ HAZOP quality is affected by the accuracy of Process Safety Information (PSI).
- ❑ **HAZOP relies critically on expertise and knowledge of the team members and team leader involved**

Basic Features of a HAZOP

HAZOP Leadership Training Course

- ❑ Structured / Systematic examination of a process or procedure, either planned or existing
- ❑ Guideword stimulated, brainstorming
- ❑ Team of subject matter experts (5-7); Led by experienced leaders
- ❑ Identify potential safety and operability problems
- ❑ Recommend actions to resolve
- ❑ Not intended to be used as a design meeting
- ❑ Attempts to answer
 - “Qs”-”Is it safe?”,
 - "Can it be operated as intended?”
 - If NOT “How can I improve the design?”

HAZOP Leadership training Course for GC

End of Chapter 3 : Introduction of HAZOP study

Technical and Safety service (Q-TS-TS)

PTT Global Chemical PCL